# PRASANTH ANNAM

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## SUMMARY

Software engineer with 10+ years of experience designing, building, and operating large-scale distributed systems, cloud-native platforms, and full-stack products at AWS, Tesla, and Western Digital. Experienced in leading teams of 5 to 10 engineers and owning systems end to end, from architecture and API design through implementation, observability, operational readiness, and launch. Combines deep expertise in Java, Spring Boot, .NET, TypeScript, React, and AWS with hands-on experience building agentic AI solutions on Amazon Bedrock. Brings a practical engineering approach shaped by work across high-volume backend services, complex web applications, data platforms, and production operations.

## TECHNICAL SKILLS

**Programming Languages:** Java, C#, Python, JavaScript, TypeScript, SQL

**Backend Frameworks:** Spring Boot, Spring Framework, ASP.NET Core, ASP.NET MVC, Node.js, Express, FastAPI, Django

**Frontend Technologies:** React, Angular, TypeScript, JavaScript, jQuery, HTML/CSS, Highcharts, D3.js

**Databases & Data:** MySQL, SQL Server, MongoDB, DynamoDB, PostgreSQL, Redis, Elasticsearch, SQL, NoSQL, Data Warehousing, ETL, Power BI

**Distributed Systems:** Microservices, Event-Driven Architecture, Kafka, RabbitMQ, REST APIs, Real-Time Systems, Asynchronous Processing, High-Volume APIs

**Cloud & DevOps:** AWS Lambda, EC2, S3, DynamoDB, RDS, API Gateway, SQS, SNS, Step Functions, CloudWatch, CloudFormation, KMS, Glue, Athena, Docker, Kubernetes, Jenkins, GitLab CI/CD, Git, Maven, Gradle

**Agentic AI:** Amazon Bedrock Agents, Knowledge Bases, Action Groups, Amazon Q Developer CLI, RAG, MCP Servers, Multi-Agent Orchestration, Privacy-Aware LLM Design, Claude

## PROFESSIONAL EXPERIENCE

**Amazon** Austin, TX

*Software Development Engineer II* Aug 2024 – Present

- Led AWS Partner Central 3.0 with **10 engineers**, moving co-sell workflows into an AWS Console React micro-frontend serving 180K+ partners within a program representing ~$5B in business value.
- Reduced module load time **57% (4.2 seconds to 1.8 seconds)** through lazy loading and LCP optimization; built 300+ metrics, 10 dashboards, and 150 alarms with automated rollback.
- Built a privacy-preserving AWS Clean Rooms pilot across **~1.2M AWS and ~2M partner accounts**, producing ~250K matched accounts for an initiative associated with ~$400M in estimated revenue.
- Led readiness for an event platform processing **~873K events/day and ~3.77M SNS publishes/day**, adding DLQs to 27 EventBridge targets to eliminate silent event loss.
- Co-designed an agentic engineering method that reduced estimated effort from **~205 to ~120 engineer-weeks** and helped drive AI adoption to 98% across ~450 engineers.

**Tesla** Austin, TX

*Senior Software Developer* Feb 2023 – Jul 2024

- Led **5 engineers** building Tesla's web-based HR operating-system services and shipped platforms used across a 300K+ employee organization.
- Built training, recognition, and profile applications for **300K+ users**, eliminating third-party dependencies and saving ~$750K annually.
- Reduced average response time **75%** and peak error rates 90% on a candidate application module through query, caching, code, and infrastructure optimization.
- Optimized **3,000+ APIs across 5 applications**, improving HR workflow turnaround time ~30%.
- Resolved high-load training-enrollment bottlenecks within **48 hours** through SQL optimization, caching, dynamic scaling, and real-time monitoring.

**Intel Corporation – Internship** Miami, FL

*Graduate Compilers Intern* May 2022 – Dec 2022

- Built Eva to automate **10+ compiler bug-hunting tasks**, cutting testing and discovery from 4 hours to 45 minutes and identifying 70+ production bugs.
- Reduced per-run compiler logs from **25 MB to 10 MB** while preserving critical diagnostics.

**Western Digital** Bangalore, India
*Senior Software Development Engineer* Aug 2017 – Jul 2021

- Led **8 engineers** building self-service BI and finance platforms that replaced proprietary tooling and delivered $4M+ in savings.
- Rebuilt a failing file-processing workflow on RabbitMQ, reducing customer complaints **~90%** and contributing $1.5M+ in savings over two years.
- Designed **~25 TB** data warehouses and centralized JIRA analytics, saving $450K+ and cutting downstream source load 45%+.
- Built reusable Angular libraries adopted across teams, reducing development time **60%**, code size 35%, and load time 40%.
- Created a dynamic charting plugin that reduced new-widget creation **80%** and saved 1,000+ engineering hours.

**Infosys** Bangalore, India
*Systems Engineer* Dec 2015 – Aug 2017

- Served as a core UI engineer for India's Goods and Services Tax application supporting **300M+ users**.
- Designed and delivered **80+ responsive pages** and role-based experiences across viewport sizes.
- Built internal tools that generated PDF documents for appraisals, announcements, and employee recognition.

## EDUCATION

**Florida International University** Miami, FL
*M.S. in Computer and Information Science, GPA: 3.97/4* Dec 2022
Relevant Courses: Machine Learning, Data Analytics

**Jawaharlal Nehru Technological University, Kakinada** Andhra Pradesh, India
*B.Tech. in Computer Science* May 2015

## ACHIEVEMENTS

- **GenAI Hackathon Winner (Amazon):** Built an AI-agent platform that turns partner problems into solution architectures, cost estimates, migration plans, and sales-ready presentations.
- **Lead with Purpose Award (Tesla):** Recognized for proactively clearing backlog items and improving application reliability.
- **Platinum, Gold, Silver, and Bronze High5 Awards (Western Digital):** Recognized repeatedly for high-impact, fast-turnaround delivery.
- **IBU Green Horn Award (Infosys):** Recognized for dependable execution on critical, tight-deadline work across teams.
- **President, Computer Society of India Student Chapter:** Led technical, programming, career-development, and student-event initiatives.